SIMATS ENGINEERING
ASSESSMENT TOOL 2

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COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0703

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the

switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or

tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then

connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department’s network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

Industry Problem-Solving Task – Answers

1. Startup Office (20 Employees)

Recommended Topology: Star Topology

Network Devices:

- 24-port Layer 2/Layer 3 Managed Switch
- Router/Firewall
- Wi-Fi Access Point
- PCs, Printers, VoIP Phones

Cable Used:

- Cat6 UTP Cable (Gigabit Ethernet)

Advantages:

- Easy to install and manage.
- If one cable fails, only that device is affected.
- Easy to add new devices.
- Centralized troubleshooting.

For a 20-person startup, the most suitable choice is the Star Topology because all computers and network devices are connected to a central switch. A 24-port Layer 2/Layer 3 managed switch is placed at the center of the network to connect workstations, printers, VoIP phones, and other devices while leaving extra ports for future expansion. Each device is connected using Cat6 UTP cables, which support Gigabit Ethernet speeds up to 100 meters and provide reliable high-speed communication. The central switch is connected to a router/firewall that manages internet access, DHCP, NAT, and network security. A Wi-Fi access point connected to the switch provides wireless connectivity for laptops and mobile devices. The major advantage of this topology is that if one cable or workstation fails, only that particular device is affected while the remaining devices continue to operate normally. The centralized design also simplifies network management, troubleshooting, and future expansion.

2. Bank Network

Recommended Topology: Full Mesh / Partial Mesh Topology

Network Devices:

- Dual Enterprise Routers
- Core Switches
- Firewalls
- Servers
- UPS and RAID Storage

Cable Used:

- Fiber Optic Cable (Single-mode or Multi-mode)

Advantages:

- Near-zero downtime.
- Multiple backup paths.
- Automatic rerouting using OSPF/BGP.
- High security and reliability.

For a bank, where uninterrupted service is critical, a Full Mesh or Partial Mesh Topology is the best solution. In this topology, all important network devices such as core routers, multilayer switches, firewalls, and servers are connected through multiple redundant links. Banks generally use dual enterprise routers (such as Cisco ASR or Juniper MX series) with BGP routing to provide redundant internet connections. The backbone uses fiber optic cables (single-mode for long distances and multi-mode for data centers) because they provide very high bandwidth, low latency, and immunity to electromagnetic interference. Additional components such as UPS systems, redundant power supplies, and RAID storage increase reliability. If one network link or device fails, routing protocols like OSPF or BGP automatically redirect traffic through another available path within milliseconds. This ensures continuous banking operations, maintains data integrity, and satisfies strict uptime requirements.

3. Small Classroom Lab

Recommended Topology: Bus Topology

Network Devices:

- Coaxial Cable (RG-58) with BNC Connectors and Terminators
- *(or)* Low-cost Unmanaged Switch with Cat5e/Cat6

Advantages:

- Very low installation cost.
- Simple setup.
- Suitable for small classroom networks.

Limitation:

- If the main cable fails, the whole network stops.
- Data collisions may occur.

For a small classroom computer laboratory with a limited budget, the Bus Topology provides a cost-effective solution. All computers share a common communication cable, reducing the need for expensive networking hardware. A traditional implementation uses RG-58 coaxial cable, BNC connectors, T-connectors, and 50-ohm terminators placed at both ends of the cable to prevent signal reflection. In a modern implementation, a low-cost 8-port or 16-port unmanaged switch with Cat5e or Cat6 cables may be used while still maintaining a simple and economical design. This topology is suitable for internet browsing, classroom activities, and basic file sharing. However, a major limitation is that if the backbone cable fails, the entire network becomes unavailable, and data collisions may occur when multiple systems communicate simultaneously. Despite these drawbacks, it remains an economical choice for small educational environments.

4. Large University Campus

Recommended Topology: Hybrid Topology

Network Devices:

- Core Routers
- Layer 3 Core Switches
- Distribution Switches
- Access Switches
- Wireless Access Points

Cable Used:

- Fiber Optic Cable (10 Gbps/40 Gbps)
- Cat6 UTP Cable

Advantages:

- High scalability.
- Redundant backbone for reliability.
- Easy to add new buildings.
- Supports wired and wireless networks.

A large university campus requires a Hybrid Topology because thousands of users across multiple buildings need both scalability and high availability. The campus backbone is connected using a fiber optic ring or partial mesh topology, ensuring redundancy between core routers and multilayer switches. High-speed single-mode fiber cables supporting 10 Gbps or 40 Gbps are commonly used to interconnect academic buildings, libraries, laboratories, hostels, and administrative offices. Each building contains a distribution switch connected to the campus core through dual fiber uplinks. Inside each building, a Star Topology is implemented where access switches connect computers, printers, and other devices using Cat6 UTP cables. Wireless access points are powered using Power over Ethernet (PoE), eliminating the need for separate electrical wiring. This three-layer hierarchical architecture consisting of Core, Distribution, and Access layers provides excellent scalability, easy maintenance, redundancy, and simplified expansion whenever new buildings are added to the campus network.

5. Company with Multiple Departments
Recommended Design: Hierarchical Network with VLANs

Network Devices:

- Layer 2 Access Switches
- Layer 3 Switch/Router
- Firewall
- Router for Internet Access

Features:

- Separate VLANs for HR, Finance, Sales, and IT.
- Inter-VLAN routing through Layer 3 Switch.
- Proper IP addressing and subnetting.
- Redundant links for reliability.

Advantages:

- Improved security.
- Reduced broadcast traffic.
- Better network performance.
- Easy expansion and management.

For a company with multiple departments such as Human Resources, Finance, Sales, and Information Technology, a hierarchical VLAN-based network architecture provides the best performance and security. Each department is connected through dedicated Layer 2 access switches, allowing communication among devices within the same department. These access switches connect to a Layer 3 switch or router, which performs inter-VLAN routing so that departments can communicate securely when required. Separate Virtual LANs (VLANs) are configured for each department to isolate broadcast traffic and improve network security Proper IP addressing and subnetting are implemented to organize the network efficiently, and redundant links provide backup communication paths in case of failures. This hierarchical design improves overall network performance, enhances security, simplifies administration, and allows the company to expand easily as additional departments and employees are added.